



Islamic University of Technology
Department of Computer Science and Engineering

Lab 11: Maximum Flow

CSE 4404: Algorithms Lab
Summer 2024-25

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Task A. Water Distribution

Time Limit: 2 seconds | Memory Limit: 256 MB

IUT's central utility plant pumps water to the campus through a network of underground pipes. The network has n junctions (including the plant itself and every building) connected by m one-way pipes, where the i -th pipe runs from junction u_i to junction v_i and can carry at most c_i litres of water per minute.

Facilities wants to know the maximum amount of water per minute that can be sustained from the central plant (junction s) all the way to the largest dormitory's tank (junction t), routing water through any combination of pipes as long as no pipe ever exceeds its capacity.

Find the maximum possible flow from s to t .

Input Format

The first line contains four integers n , m , s , and t , the number of junctions, the number of pipes, the source junction, and the target junction.

Each of the next m lines contains three integers u_i , v_i , and c_i , describing a one-way pipe from u_i to v_i with capacity c_i litres per minute. There may be more than one pipe between the same ordered pair of junctions, and a pipe from u to v does not imply a pipe from v to u .

Output Format

Print a single integer, the maximum flow from s to t .

Constraints

- $2 \leq n \leq 300$
- $1 \leq m \leq 1500$
- $1 \leq s, t \leq n, s \neq t$
- $1 \leq c_i \leq 10^9$

Examples

Sample Input	Sample Output
4 5 1 4 1 2 3 1 3 2 2 3 5 2 4 2 3 4 3	5
3 2 1 3 1 2 4 2 3 7	4
3 1 1 3 1 2 5	0

Task B. Minimum Sabotage

Time Limit: 2 seconds | Memory Limit: 256 MB

IUT's IT security team is stress-testing the campus network. The network has n machines (numbered 1 to n) connected by m one-way links, where the i -th link runs from machine u_i to machine v_i and can carry at most c_i units of bandwidth.

The team wants to know the minimum total bandwidth of links that must be completely severed so that *no* data can travel from the server room (machine s) to the target lab (machine t), even indirectly through several hops.

After finding this minimum-bandwidth set of links to sever, the team also needs the resulting split of the network: which machines remain reachable from the server room, and which machines get cut off. Help them find the minimum total bandwidth to sever, along with the set of machines that stay reachable from the server room afterwards.

It is guaranteed that the input data has exactly one way to achieve the minimum total severed bandwidth, so the resulting split is unambiguous.

Input Format

The first line contains four integers n , m , s , and t , the number of machines, the number of links, the server room, and the target lab.

Each of the next m lines contains three integers u_i , v_i , and c_i , describing a one-way link from u_i to v_i with bandwidth c_i . There may be more than one link between the same ordered pair of machines, and a link from u to v does not imply a link from v to u .

Output Format

On the first line, print a single integer, the minimum total bandwidth that must be severed.

On the second line, print an integer k followed by k space-separated integers, the machines that remain reachable from the server room (including the server room itself) after severing that minimum-bandwidth set of links, listed in increasing order.

Constraints

- $2 \leq n \leq 300$
- $1 \leq m \leq 1500$
- $1 \leq s, t \leq n, s \neq t$
- $1 \leq c_i \leq 10^9$
- The minimum severed bandwidth is achieved by exactly one split of the machines into a reachable side and a cut-off side.

Examples

Sample Input	Sample Output
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4 3 1 4 1 2 100 2 3 1 3 4 100	1 2 1 2
5 5 1 5 1 2 50 1 3 50 2 4 50 3 4 50 4 5 8	8 4 1 2 3 4

Marks Distribution

Task	Marks
Task A	40%
Task B	60%

Practice Problems

Problem	Link
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